

# Data-Driven Q-Learning for Consensus and Multi-Consensus Control of Discrete-Time Linear Multi-Agent Systems

Md Nur-A-Adam Dony<sup>a</sup>, Syed Ali Asad Rizvi<sup>a,\*</sup>

<sup>a</sup>*Department of Electrical and Computer Engineering, Tennessee Technological University, Cookeville, 38505, TN, USA*

---

## Abstract

This paper presents data-driven Q-learning algorithms for consensus and multi-consensus control of discrete-time linear multi-agent systems. Both state feedback and output feedback formulations learn the gains of a predictor-based consensus protocol directly from measured data, without knowledge of the agent dynamics. The state feedback algorithm recovers both gains from a single agent's state-input trajectory, with convergence to the discrete algebraic Riccati equation solution established rigorously. The output feedback extension learns the input-coupling gain and reconstructs the inter-agent disagreement feedback from input-output data alone, eliminating full state measurement while preserving the consensus properties of the underlying design. Every example is validated under both feedback types, and the state feedback and output feedback deployments are shown to produce trajectories that coincide to machine precision, as the theory predicts. A numerical robustness analysis demonstrates the advantage of the data-driven gains under parametric model uncertainty, and simulations on multi-consensus, formation control, and vehicle-platoon scenarios validate the results against the model-based benchmark.

**Keywords:** Multi-agent systems, Consensus control, Q-learning, Output feedback, Data-driven control, Discrete-time systems

---

---

\*Corresponding author.

*Email addresses:* mdony42@tntech.edu (Md Nur-A-Adam Dony), srizvi@tntech.edu (Syed Ali Asad Rizvi)

## 1. Introduction

Consensus control of networked multi-agent systems (MAS) is a fundamental problem in distributed control theory, with applications in formation control of unmanned vehicles [1, 2, 3], synchronization of complex networks [4, 5], distributed filtering [6], and opinion dynamics in social networks [7]. The central objective is to design distributed control laws that drive all agents toward common trajectories using only local information exchanged over a communication graph. A well-known challenge distinguishes the discrete-time and continuous-time settings. For continuous-time MAS, arbitrarily fast convergence to consensus can be achieved simply by increasing the scalar coupling gain [8, 9]. In sharp contrast, for discrete-time systems, the coupling gain is constrained to be small—typically inversely proportional to the largest eigenvalue of the Laplacian matrix—to maintain closed-loop stability [10, 11, 12]. This constraint fundamentally limits the achievable convergence rate. Recently, Cacace et al. [13] proposed a predictor-based consensus protocol for discrete-time LTI MAS that overcomes this limitation through an implicit feedback law involving both the state and input of neighboring agents, recovering arbitrary convergence rate via multiple consensus steps per sampling period. The analysis extends to weakly connected digraphs through almost equitable partitions [14], and the design requires solving a single DARE for two gain matrices  $F$  and  $G$ . However, this requires exact knowledge of  $(A, B)$ , which may not be available in practice due to modeling errors, parametric variations, or discretization approximations [15, 3, 16]. Reinforcement learning (RL) offers a model-free alternative [17]. Q-learning solves the LQR problem without system knowledge [18] using policy iteration (PI) or value iteration (VI), and output feedback RL reconstructs the state from input-output data [17, 19]. For MAS, Zhang et al. [20] developed RL-based optimal synchronization, and related data-driven methods include safe RL for discrete event systems [21], battery balancing [22], and Koopman-based control [23]. However, these methods address the standard explicit protocol  $u_i = -\kappa \sum_j F(x_i - x_j)$ . The predictor-based protocol of [13] has a fundamentally different implicit structure:  $u_i$  depends on both states *and inputs* of neighbors through  $G$ , requiring a matrix inversion  $(I + \tilde{\mathcal{L}} \otimes G)^{-1}$  to resolve the coupling, which prevents direct application of existing RL-based MAS methods. Integrating model-free RL with predictor-based consensus protocols has not been addressed.

**Novelty and context:** In this paper, we learn the consensus gains of the predictor protocol of [13] directly from data. The entry point is structural: the consensus DARE coincides with the LQR Riccati equation for  $R =$

$I_m$ , so single-agent Q-learning can extract both gains  $F$  and  $G$  without knowledge of  $(A, B)$ . This equivalence is not itself the contribution; a direct composition of established Q-learning [18, 17, 19] with the protocol fails on three counts that the paper resolves: (a) the protocol is implicit in  $u_i$  through  $G(u_i - u_j)$ , which blocks existing RL-based MAS methods [20] and forces a graph-independent single-agent formulation; (b) output-feedback Q-learning returns the absolute regulator  $-K^*x_i$ , which drives each agent to the origin, so consensus instead requires reconstructing the inter-agent disagreement  $F(x_i - x_j)$  from input-output data (Proposition 1); and (c) only *exact* recovery of the DARE solution preserves the arbitrary convergence rate of [13]. The novelty therefore lies in the linkage between two established but previously unconnected areas—data-driven LQR reinforcement learning and predictor-based discrete-time consensus—and in the new capability this linkage enables. Table 1 summarizes these distinctions.

Impact and insight: The resulting framework removes the model requirement from a consensus design whose principal appeal is its arbitrary convergence rate, extending its applicability to systems with uncertain, drifting, or poorly discretized dynamics. The insight that both predictor gains reside in the LQR Q-function—with  $G$  obtained from the input block that the absolute regulator discards—and that inter-agent disagreement feedback can be reconstructed purely from input-output filter states, is transferable to other implicit or input-coupled distributed control structures.

The specific contributions are:

- (i) A single-agent learning architecture that decouples the computation of  $(F, G)$  from the network topology, since these gains depend only on  $(A, B, Q)$ ; the learning cost is thus independent of the number of agents.
- (ii) An output feedback extension that learns the input-coupling gain  $G$  and reconstructs the disagreement feedback  $F(x_i - x_j)$  from input-output data alone (Proposition 1), with no state observer or system identification.
- (iii) Rigorous convergence proofs showing the Q-learned gains satisfy the DARE and inherit all consensus and multi-consensus guarantees, including arbitrary convergence rate; this includes an extension of the stability condition of [13] to the positive semi-definite weights  $Q = C^\top Q_y C$  that the output feedback formulation necessitates (Remark 1).
- (iv) Quantitative robustness analysis demonstrating that Q-learned gains maintain superior stability margins and lower control costs compared to model-based gains under parametric uncertainty.

- (v) Comprehensive simulation validation in which *every* scenario—multi-consensus, formation control, robustness, and a vehicle platoon—is executed under both state and output feedback, with the two deployments shown to coincide to machine precision, benchmarked against the model-based design of [13].

The paper follows the journal structure. Section 2 (Methods) presents the preliminaries, the problem formulation, and the state feedback and output feedback Q-learning algorithms with their convergence guarantees. Section 3 (Results) reports the simulation studies. Section 4 (Discussion) analyzes computational complexity, scalability, communication requirements, robustness, and limitations. Section 5 concludes.

Table 1: Comparison of the proposed framework with existing methods.

| Feature                          | [13] | [20] | [18] | [19] | <b>Proposed</b> |
|----------------------------------|------|------|------|------|-----------------|
| Model-free                       | No   | Yes  | Yes  | Yes  | <b>Yes</b>      |
| Output feedback                  | No   | No   | No   | Yes  | <b>Yes</b>      |
| Arbitrary conv. rate             | Yes  | No   | N/A  | N/A  | <b>Yes</b>      |
| Single-agent learning            | N/A  | No   | Yes  | Yes  | <b>Yes</b>      |
| Implicit $G(u_i - u_j)$ coupling | Yes  | No   | No   | No   | <b>Yes</b>      |
| Multi-consensus                  | Yes  | No   | No   | No   | <b>Yes</b>      |

## 2. Methods

This section formulates the model-free consensus gain learning problem and develops the two proposed algorithms. The methodological novelty is twofold: a structural equivalence that lets a *single-agent* LQR Q-function deliver *both* predictor gains  $F$  and  $G$  (state feedback), and a disagreement-reconstruction result that realizes the predictor protocol from input-output data alone (output feedback). Neither is available from standard LQR Q-learning [18, 17, 19], which produces a single absolute regulator gain for an isolated system and is not connected to a consensus protocol.

### 2.1. Preliminaries and Problem Formulation

#### 2.1.1. Graph Theory

Consider a digraph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  with  $|\mathcal{V}| = N$  nodes. The in-neighbor set of node  $i$  is  $\mathcal{N}_i = \{j \in \mathcal{V} : (j, i) \in \mathcal{E}\}$  with in-degree  $d_i = |\mathcal{N}_i|$ . The Laplacian matrix is  $\mathcal{L} = \mathcal{D} - \mathcal{A}$ , where  $\mathcal{D} = \text{diag}(d_1, \dots, d_N)$  and  $\mathcal{A}$  is the adjacency

Table 2: Notation used throughout the paper.

| Symbol                               | Description                        | Symbol                             | Description                                                           |
|--------------------------------------|------------------------------------|------------------------------------|-----------------------------------------------------------------------|
| $\mathbb{R}, \mathbb{C}, \mathbb{N}$ | Real, complex, natural numbers     | $\text{vecs}(H)$                   | Vectorization of symmetric $H$                                        |
| $\Re(z)$                             | Real part of $z \in \mathbb{C}$    | $\mathcal{L}, \tilde{\mathcal{L}}$ | Laplacian, modified Laplacian                                         |
| $I_n$                                | Identity of size $n$               | $\bar{r}$                          | $\min_{\lambda \in \sigma(\mathcal{L}) \setminus \{0\}} \Re(\lambda)$ |
| $\mathbb{1}_N$                       | All-ones vector of length $N$      | $F, G$                             | Consensus gain matrices                                               |
| $\sigma(A)$                          | Spectrum of $A$                    | $H, \tilde{H}$                     | State/output FB Q-matrices                                            |
| $\rho(A)$                            | Spectral radius $\max  \sigma(A) $ | $K, \bar{K}$                       | State/output FB policies                                              |
| $A \otimes B$                        | Kronecker product                  | $x^+$                              | $x(t+1)$ , successor                                                  |
| $\text{col}_i(x_i)$                  | $(x_1^\top, \dots, x_N^\top)^\top$ |                                    |                                                                       |

matrix. By construction,  $\lambda_1 = 0 \in \sigma(\mathcal{L})$  with eigenvector  $\mathbb{1}_N$ . A digraph has *one reach* if it is weakly connected with a set of root nodes from which all others are reachable; in this case  $\lambda_1 = 0$  is simple and  $\Re(\lambda_k) > 0$  for  $k \geq 2$  [24]. We define

$$\bar{r} = \min_{\lambda \in \sigma(\mathcal{L}) \setminus \{0\}} \Re(\lambda), \quad (1)$$

which determines the minimum coupling gain needed for consensus. For general weakly connected digraphs with  $\mu$  reaches,  $\lambda_1 = 0$  has multiplicity  $\mu$  [24], and the coarsest almost equitable partition (AEP)  $\pi^*$  [14] induces the block-triangular Laplacian structure

$$\mathcal{L} = \begin{pmatrix} \text{diag}_{k=1}^\mu(\mathcal{L}_k) & 0 \\ \text{row}_{k=1}^\mu(\mathcal{M}_k) & \mathcal{M} \end{pmatrix}, \quad (2)$$

where each  $\mathcal{L}_k$  has a simple zero eigenvalue and  $\sigma(\mathcal{M}) \subset \{z \in \mathbb{C} : \Re(z) > 0\}$ .

### 2.1.2. System Model and Consensus Problem

Consider  $N$  identical discrete-time LTI agents with dynamics

$$x_i(t+1) = Ax_i(t) + Bu_i(t), \quad (3)$$

$$y_i(t) = Cx_i(t), \quad (4)$$

where  $x_i \in \mathbb{R}^n$ ,  $u_i \in \mathbb{R}^m$ ,  $y_i \in \mathbb{R}^p$ , and common matrices  $A \in \mathbb{R}^{n \times n}$ ,  $B \in \mathbb{R}^{n \times m}$ ,  $C \in \mathbb{R}^{p \times n}$ . The collective dynamics is

$$\mathbf{x}^+ = (I_N \otimes A)\mathbf{x}(t) + (I_N \otimes B)\mathbf{u}(t), \quad (5)$$

where  $\mathbf{x} = \text{col}_{i=1}^N(x_i) \in \mathbb{R}^{nN}$  and  $\mathbf{u} = \text{col}_{i=1}^N(u_i) \in \mathbb{R}^{mN}$ .

*Assumption 1* (Controllability). The pair  $(A, B)$  is controllable.

*Assumption 2* (Observability). The pair  $(A, C)$  is observable.

*Assumption 3* (Graph Connectivity). The communication digraph  $\mathcal{G}$  is weakly connected.

The consensus problem (for  $\mu = 1$ ) requires  $x_i(t) \rightarrow A^t x_s(0)$  for all  $i$ . The multi-consensus problem (for general  $\mu$ ) requires agents in each cell of  $\pi^\star$  to converge to a common trajectory [13].

### 2.1.3. Predictor-Based Consensus Protocol

The predictor-based consensus protocol from [13] for each agent  $i$  is

$$u_i = -\kappa_i \sum_{j \in \mathcal{N}_i} [F(x_i - x_j) + G(u_i - u_j)], \quad (6)$$

where  $\kappa_i > 0$ ,  $F \in \mathbb{R}^{m \times n}$ , and  $G \in \mathbb{R}^{m \times m}$ . The term “predictor-based” refers to the fact that the protocol contains neighboring input differences  $u_i - u_j$  in addition to state differences  $x_i - x_j$ . Since the one-step disagreement evolution satisfies

$$x_i^+ - x_j^+ = A(x_i - x_j) + B(u_i - u_j), \quad (7)$$

and the consensus gains satisfy  $F = B^\top P A$  and  $G = B^\top P B$ , the disagreement feedback in (6) can be written as  $F(x_i - x_j) + G(u_i - u_j) = B^\top P(x_i^+ - x_j^+)$ . That is, the protocol implicitly accounts for the one-step evolution of the disagreement dynamics. This predictor-like input coupling modifies each disagreement mode through the closed-loop matrix (9), which is the mechanism that removes the small-gain restriction of standard discrete-time consensus protocols [10]. Note that (6) is implicit:  $u_i$  appears on both the left-hand side and inside the summation through  $G(u_i - u_j)$ . This coupling is resolved globally through the matrix inversion in the centralized form (8), and locally through the iterative  $\gamma$ -step algorithm (12)–(13). With the modified Laplacian  $\tilde{\mathcal{L}} = \text{diag}(\kappa_1, \dots, \kappa_N) \mathcal{L}$ , the centralized solution is

$$\mathbf{u} = -(I_{mN} + \tilde{\mathcal{L}} \otimes G)^{-1} (\tilde{\mathcal{L}} \otimes F) \mathbf{x}. \quad (8)$$

The closed-loop dynamics is  $\mathbf{x}^+ = \Theta \mathbf{x}$ , with spectral decomposition [13, Lemma 3.1] giving  $\sigma(\Theta) = \bigcup_{\tilde{\lambda}_k \in \sigma(\tilde{\mathcal{L}})} \sigma(\tilde{A}_k)$  where

$$\tilde{A}_k = A - B \left( \frac{1}{\tilde{\lambda}_k} I_m + G \right)^{-1} F. \quad (9)$$

For any  $Q = Q^\top > 0$ , the model-based design of [13, Thm. 3.1] takes  $P = P^\top \succ 0$  solving the DARE

$$A^\top P A - P - A^\top P B (I_m + B^\top P B)^{-1} B^\top P A + Q = 0, \quad (10)$$

and sets

$$F = B^\top P A, \quad G = B^\top P B, \quad (11)$$

for which the closed-loop modes  $\bar{A}_k$  in (9) are Schur stable provided  $\kappa_i \bar{r} > 1$  for all  $i$ . The convergence rate is regulated by  $Q$  [13, Remark 3.2]. For practical deployment, a distributed  $\gamma$ -step algorithm approximates (8) [13]:

$$v_i(t, 0) = -\kappa_i(I_m + \kappa_i d_i G)^{-1} \sum_{j \in \mathcal{N}_i} F(x_i - x_j), \quad (12)$$

$$v_i(t, h+1) = v_i(t, 0) + \kappa_i(I_m + \kappa_i d_i G)^{-1} \sum_{j \in \mathcal{N}_i} G v_j(t, h), \quad (13)$$

for  $h = 0, \dots, \gamma - 1$ , with  $u_i(t) = v_i(t, \gamma)$ .

*Remark 1* (Positive semi-definite weights). Theorem 3.1 of [13] is stated for  $Q \succ 0$ , but the output feedback formulation of Section 2.3 necessarily employs  $Q = C^\top Q_y C \succeq 0$ , since only measured outputs can be weighted. The stability conclusion persists when  $(A, \sqrt{Q})$  is observable. Indeed, the Lyapunov computation in the proof of [13, Thm. 3.1] yields, for each nonzero  $\bar{\lambda}_k$  with  $\Re(\bar{\lambda}_k) > 1$ ,

$$\bar{A}_k^* P \bar{A}_k - P = -Q - \frac{\Re(\bar{\lambda}_k)}{|\bar{\lambda}_k|^2} F^\top S_k^* S_k F \preceq 0, \quad S_k = \left( \frac{1}{\bar{\lambda}_k} I_m + G \right)^{-1}.$$

Suppose  $\bar{A}_k v = \mu v$  with  $|\mu| \geq 1$ ,  $v \neq 0$ . Multiplying the identity above by  $v^*$  and  $v$  gives  $(|\mu|^2 - 1) v^* P v = -v^* Q v - c_k \|S_k F v\|^2$  with  $c_k > 0$  and  $P \succ 0$ , so both sides must vanish:  $|\mu| = 1$ ,  $\sqrt{Q} v = 0$ , and  $F v = 0$ . But  $F v = 0$  implies  $\bar{A}_k v = A v = \mu v$ , i.e.,  $(\mu, v)$  is an unobservable unit-circle mode of  $(A, \sqrt{Q})$ , contradicting observability. Hence every  $\bar{A}_k$  is Schur, and all consensus and multi-consensus conclusions of [13] carry over verbatim to  $Q = C^\top Q_y C$  under Assumption 2 on  $(A, \sqrt{Q})$ .

**Problem Statement.** The DARE (10) and the gain formulas (11) require explicit knowledge of  $(A, B)$ , which motivates the following data-driven formulation.

**Problem 1** (Model-free consensus gain learning). *Given a performance weight  $Q = C^\top Q_y C$  with  $(A, \sqrt{Q})$  observable and measured trajectories*

$$\mathcal{D} = \{(x_k, u_k)\}_{k=0}^L \text{ (state feedback) or } \{(u_k, y_k)\}_{k=0}^L \text{ (output feedback)}, \quad (14)$$

with the dynamics  $(A, B, C)$  unknown, find gains  $F \in \mathbb{R}^{m \times n}$  and  $G \in \mathbb{R}^{m \times m}$  equal to the model-based gains (11),

$$F = B^\top P A, \quad G = B^\top P B, \quad P = P^\top \succ 0 \text{ solving (10)}, \quad (15)$$

such that the protocol (6) renders the network (multi-)consensus,

$$\lim_{t \rightarrow \infty} (x_i(t) - x_j(t)) = 0, \quad \forall i, j \text{ in a common cell of } \pi^*, \quad (16)$$

using only the data  $\mathcal{D}$  and the weight  $Q_y$ , without  $A$ ,  $B$ , or  $C$ .

#### 2.1.4. $Q$ -Learning for the LQR Problem

We briefly recall the standard LQR  $Q$ -learning framework [18, 17, 19] on which our development builds; it is included only for notation and is not part of the contribution. For a single agent  $x_{k+1} = Ax_k + Bu_k$  with cost  $V_K(x_k) = \sum_{i=k}^{\infty} (x_i^\top Qx_i + u_i^\top Ru_i)$  under  $u_i = Kx_i$ , the  $Q$ -function is quadratic,

$$Q_K(x_k, u_k) = z_k^\top H z_k, \quad z_k = \begin{bmatrix} x_k \\ u_k \end{bmatrix}, \quad H = \begin{bmatrix} Q + A^\top P A & A^\top P B \\ B^\top P A & R + B^\top P B \end{bmatrix}, \quad (17)$$

and the optimal policy is  $u_k^* = -K^* x_k$  with  $K^* = (H_{uu}^*)^{-1} H_{ux}^*$  [18]. The Bellman equation

$$z_k^\top H z_k = x_k^\top Q x_k + u_k^\top R u_k + z_{k+1}^\top H z_{k+1} \quad (18)$$

is model-free—it involves only the measurable quantities  $(x_k, u_k, x_{k+1})$  and known  $(Q, R)$ . Vectorizing  $H$  as  $\bar{H} = \text{vecs}(H)$  yields the linear regression

$$\bar{H}^\top (\bar{z}_k - \bar{z}_{k+1}) = x_k^\top Q x_k + u_k^\top R u_k, \quad (19)$$

solved by least squares under a rank (excitation) condition. PI requires a stabilizing initial  $K^0$ ; VI does not [17]. Exploration noise added to  $u_k = -Kx_k + e_k$  does not bias the solution [18, 19]. The proposed methodology is illustrated in Fig. 1.

#### 2.2. State Feedback $Q$ -Learning for Consensus Control

The technical development is twofold: a structural equivalence that lets a *single-agent* LQR  $Q$ -function deliver *both* predictor gains  $F$  and  $G$ , and a proof that the learned gains coincide *exactly* with the DARE solution so that the consensus guarantees of [13] are inherited rather than approximated.

##### 2.2.1. Connecting the DARE to the LQR $Q$ -Function

Comparing the consensus DARE (10) with the standard LQR DARE

$$A^\top P A - P + Q - A^\top P B (R + B^\top P B)^{-1} B^\top P A = 0, \quad (20)$$

setting  $R = I_m$  yields (10) exactly. The value  $R = I_m$  is not a design choice but is *forced*: it is the unique weight for which the LQR Riccati equation reproduces



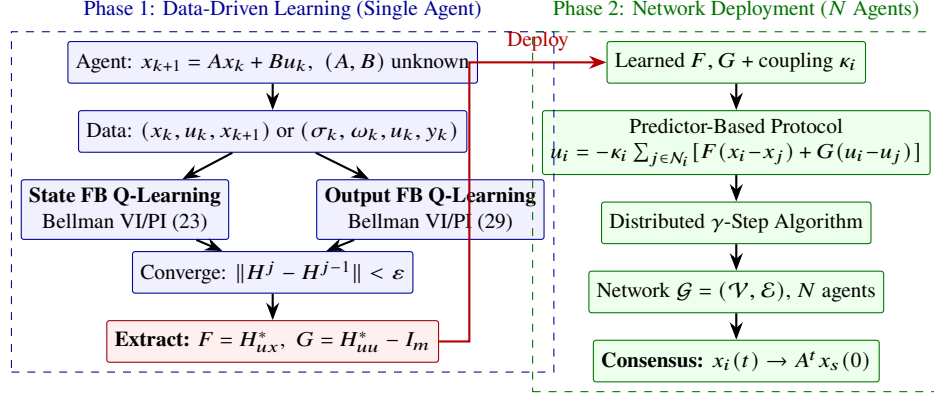


Figure 1: Proposed data-driven methodology. Phase 1: offline Q-learning on a single agent to extract consensus gains  $F$  and  $G$ . Phase 2: deployment in the predictor-based protocol across the multi-agent network.

the consensus DARE of [13], and hence the unique weight for which the learned gains coincide with (11). The consequence is that the consensus gains can be read off the LQR Q-function without ever forming or solving the DARE, and without any knowledge of  $(A, B)$ .

**Lemma 1.** *Let  $H^*$  be the optimal H-matrix of the LQR problem with weights  $(Q, I_m)$ . Then*

$$F = H_{ux}^*, \quad G = H_{uu}^* - I_m. \quad (21)$$

*Proof.* Under the optimal value  $V^*(x_k) = x_k^\top P^* x_k$  and  $R = I_m$ , the Q-function (17) expands as

$$\begin{aligned} Q^*(x_k, u_k) &= x_k^\top Q x_k + u_k^\top u_k + (Ax_k + Bu_k)^\top P^* (Ax_k + Bu_k) \\ &= \underbrace{\begin{bmatrix} x_k \\ u_k \end{bmatrix}^\top \begin{bmatrix} Q + A^\top P^* A & A^\top P^* B \\ B^\top P^* A & I_m + B^\top P^* B \end{bmatrix} \begin{bmatrix} x_k \\ u_k \end{bmatrix}}_{H^*}. \end{aligned}$$

Reading off the lower blocks of  $H^*$ ,

$$\begin{aligned} H_{ux}^* &= B^\top P^* A = F, \\ H_{uu}^* &= I_m + B^\top P^* B = I_m + G, \end{aligned}$$

gives  $F = H_{ux}^*$  and  $G = H_{uu}^* - I_m$ . □

Whereas standard Q-learning uses only  $H_{ux}^*$  to form the regulator  $K^* = (H_{uu}^*)^{-1}H_{ux}^*$ , Lemma 1 shows that the *two* predictor gains needed by (6) are both present in  $H^*$ , with  $G$  obtained from the input block  $H_{uu}^*$  that the absolute regulator discards.

### 2.2.2. Model-Free Bellman Equation for Consensus Gains

Lemma 1 reduces the consensus gain learning problem to an LQR Q-learning problem with  $R = I_m$ , run on a single agent. Setting  $R = I_m$  in (18) gives

$$z_k^\top H z_k = x_k^\top Q x_k + u_k^\top u_k + z_{k+1}^\top H z_{k+1}, \quad (22)$$

which involves only the measurable pairs  $(x_k, u_k, x_{k+1})$  and the user-specified  $Q$ . Vectorizing yields

$$\bar{H}^\top (\bar{z}_k - \bar{z}_{k+1}) = x_k^\top Q x_k + u_k^\top u_k. \quad (23)$$

Collecting  $L \geq l(l+1)/2$  samples (where  $l = n + m$ ) under probing input  $u_k = -K^j x_k + e_k$  yields the batch equation  $\Phi \bar{H} = \Psi$ , solved by least squares if  $\text{rank}(\Phi) = l(l+1)/2$ . PI solves for  $H^j$  and updates  $K^{j+1} = (H_{uu}^j)^{-1}H_{ux}^j$ ; VI solves

$$(\bar{H}^{j+1})^\top \bar{z}_k = x_k^\top Q x_k + u_k^\top u_k + (\bar{H}^j)^\top \bar{z}_{k+1}. \quad (24)$$

Upon convergence, the consensus gains are extracted as  $F = H_{ux}^*$ ,  $G = H_{uu}^* - I_m$ . The decoupling here is the resolution of the implicit coupling noted in Section 1: because the DARE depends only on  $(A, B, Q)$  and not on  $\bar{L}$ , the gains are learned on one agent and then deployed across all  $N$  agents with their coupling gains  $\kappa_i$ , so the learning cost is independent of network size. This is what makes the implicit MAS problem tractable by single-agent learning.

### 2.2.3. Convergence and Consensus Guarantees

**Theorem 1** (State Feedback Convergence and Consensus). *Let Assumptions 1 and 3 hold, and let  $(A, \sqrt{Q})$  be observable. Then:*

- (i) *The Q-learning iterations converge:  $H^j \rightarrow H^*$  as  $j \rightarrow \infty$ .*
- (ii) *The extracted gains  $F = H_{ux}^*$  and  $G = H_{uu}^* - I_m$  satisfy (11).*
- (iii) *The protocol (6) with Q-learned gains achieves (multi-)consensus if  $\kappa_i \bar{r} > 1$ .*
- (iv) *Exploration noise does not bias the learned H-matrix.*

**Proof. (i) Convergence.** With  $R = I_m$ , policy iteration alternates policy evaluation and improvement,

$$\begin{aligned} P^j &= Q + (K^j)^\top K^j + (A - BK^j)^\top P^j (A - BK^j), \\ K^{j+1} &= (I_m + B^\top P^j B)^{-1} B^\top P^j A, \end{aligned}$$

which is Hwer's algorithm:  $P^j$  decreases monotonically to the unique stabilizing solution  $P^*$  of (10), with  $A - BK^j$  Schur at every step [25]. Value iteration replaces the Lyapunov solve by the Riccati recursion

$$P^{j+1} = Q + A^\top P^j A - A^\top P^j B (I_m + B^\top P^j B)^{-1} B^\top P^j A,$$

which converges to  $P^*$  from any  $P^0 \geq 0$  when  $(A, \sqrt{Q})$  is observable [17, 19]. In either case the reconstructed matrix converges,

$$H^j = \begin{bmatrix} Q + A^\top P^j A & A^\top P^j B \\ B^\top P^j A & I_m + B^\top P^j B \end{bmatrix} \longrightarrow H^*.$$

**(ii) Exact gain recovery.** By Lemma 1, the converged blocks are

$$F = H_{ux}^* = B^\top P^* A, \quad G = H_{uu}^* - I_m = B^\top P^* B,$$

and since  $P^*$  solves the DARE (10), these coincide *exactly* with the model-based gains (11)—an equality, not an approximation. **(iii) Consensus.** Because  $F$  and  $G$  satisfy (11), the design condition of [13, Thm. 3.1], extended to  $Q = C^\top Q_y C \succeq 0$  by Remark 1, applies: the modes

$$\bar{A}_k = A - B \left( \frac{1}{\lambda_k} I_m + G \right)^{-1} F$$

in (9) are Schur stable for every nonzero  $\bar{\lambda}_k$  whenever  $\kappa_i \bar{r} > 1$ . For  $\mu > 1$  reaches, the block-triangular structure (2) drives the agents in each cell of  $\pi^*$  to a common trajectory [13, Thm. 4.1]. **(iv) Noise immunity.** Under  $u_k = -K^j x_k + e_k$  the successor state is

$$x_{k+1} = (A - BK^j)x_k + B e_k,$$

so  $e_k$  enters  $z_k$  and  $z_{k+1}$  symmetrically and cancels in (23); the least-squares solution therefore equals the noise-free one whenever  $\text{rank}(\Phi) = l(l+1)/2$  [18, 19].  $\square$

The complete procedure is summarized in Algorithm 1.

### 2.3. Output Feedback Q-Learning for Consensus Control

When only  $y_i = Cx_i$  is available, the gains must be learned from input-output data. The state-reconstruction machinery used below—the parameterization, its rank condition, the output-feedback Q-function, the steady-state equivalence, and the convergence and noise-immunity proofs—is the discrete-time output-feedback

---

**Algorithm 1** State Feedback Q-Learning for Consensus Gains

---

**Require:** System order  $n$ , input dimension  $m$ ; cost matrix  $Q = C^\top Q_y C$ ; tolerance  $\varepsilon > 0$ .

**Ensure:** Consensus gains  $F \in \mathbb{R}^{m \times n}$ ,  $G \in \mathbb{R}^{m \times m}$ .

- 1: Set  $R = I_m$ ,  $l = n + m$ ,  $N_p = l(l + 1)/2$ ; define selection matrices  $E_x = \begin{bmatrix} I_n & 0 \end{bmatrix} \in \mathbb{R}^{n \times l}$ ,  $E_u = \begin{bmatrix} 0 & I_m \end{bmatrix} \in \mathbb{R}^{m \times l}$ .
  - 2: Initialize  $K^0 \in \mathbb{R}^{m \times n}$  (stabilizing for PI; zero for VI).
  - 3: Initialize  $\bar{H}^0 = \text{vecs}(I_l) \in \mathbb{R}^{N_p}$ ,  $j \leftarrow 0$ .
  - 4: **repeat**
  - 5:   — **Data Collection** —
  - 6:   **for**  $k = 0, 1, \dots, L - 1$  where  $L \geq N_p$  **do**
  - 7:     Apply  $u_k = -K^j x_k + e_k$ , measure  $x_{k+1}$
  - 8:     Form  $z_k = [x_k^\top \ u_k^\top]^\top \in \mathbb{R}^l$
  - 9:     Compute  $\bar{z}_k \in \mathbb{R}^{N_p}$  from unique elements of  $z_k z_k^\top$
  - 10:    Compute stage cost  $r_k = x_k^\top Q x_k + u_k^\top u_k$
  - 11:   **end for**
  - 12:   — **Least Squares Solve** —
  - 13:    **PI:** Build  $\Phi \in \mathbb{R}^{L \times N_p}$ : row  $k = (\bar{z}_k - \bar{z}_{k+1})^\top$ ;  $\Psi \in \mathbb{R}^L$ : entry  $k = r_k$ ;  
      solve  $\Phi \bar{H}^j = \Psi$
  - 14:    **VI:** Build  $\Phi \in \mathbb{R}^{L \times N_p}$ : row  $k = \bar{z}_k^\top$ ;  $\Psi \in \mathbb{R}^L$ : entry  $k = r_k + (\bar{H}^j)^\top \bar{z}_{k+1}$ ;  
      solve  $\Phi \bar{H}^{j+1} = \Psi$
  - 15:    — **Policy Update** —
  - 16:    Reshape  $\bar{H}^j \rightarrow H^j \in \mathbb{R}^{l \times l}$  (symmetric); extract  $H_{ux}^j = E_u H^j E_x^\top$ ,  $H_{uu}^j = E_u H^j E_u^\top$
  - 17:    Update  $K^{j+1} = (H_{uu}^j)^{-1} H_{ux}^j$ ,  $j \leftarrow j + 1$
  - 18: **until**  $\|\bar{H}^j - \bar{H}^{j-1}\| < \varepsilon$
  - 19:   — **Extract Consensus Gains** —
  - 20:  $F = H_{ux}^*$ ,  $G = H_{uu}^* - I_m$
-

Q-learning of Rizvi and Lin [19], which we cite rather than reproduce. That framework, however, learns the *absolute* regulator  $-K^*x_i$  and is not connected to a consensus protocol. The technical development of this section, given in Proposition 1, is that the learned output-feedback blocks reconstruct the inter-agent *disagreement* feedback required by (6).

### 2.3.1. State Parameterization and Output-Feedback Q-Function

Under Assumption 2, the state admits the input-output parameterization of [19, Thm. 2.1],

$$x_k = W_u \sigma_k + W_y \omega_k + (A + LC)^k x_0, \quad (25)$$

where  $L$  makes  $A + LC$  Schur,  $W_u \in \mathbb{R}^{n \times mn}$ ,  $W_y \in \mathbb{R}^{n \times pn}$ , and  $\sigma_k, \omega_k$  are the states of filters driven by the past inputs and outputs,

$$\sigma_{k+1}^i = \mathcal{A} \sigma_k^i + \mathcal{B} u_k^i, \quad \omega_{k+1}^i = \mathcal{A} \omega_k^i + \mathcal{B} y_k^i, \quad (26)$$

with  $\mathcal{A}$  a companion matrix and  $\mathcal{B} = [1 \ 0 \ \cdots \ 0]^\top$ . Placing the eigenvalues of  $\mathcal{A}$  at the origin yields the dead-beat form  $x_k = W_u \bar{u}_{k-1, k-n} + W_y \bar{y}_{k-1, k-n}$  for  $k \geq n$  [19, Thm. 2.2], and  $W = [W_u \ W_y]$  has full row rank whenever  $A$  and  $A + LC$  share no eigenvalue [19, Thm. 2.4]; in the dead-beat case this requires only the state dimension  $n$  and that  $A$  have no zero eigenvalue (otherwise the general parameterization (25) is used with nonzero filter eigenvalues [19, Rem. 2.2]). Substituting (25) (in steady state) into (17) yields the output-feedback Q-function [19, Eqs. (2.19)–(2.20)]

$$Q_K = \tilde{z}_k^\top \tilde{H} \tilde{z}_k, \quad \tilde{z}_k = [\sigma_k^\top \ \omega_k^\top \ u_k^\top]^\top, \quad \tilde{l} = mn + pn + m, \quad (27)$$

whose lower blocks are

$$\tilde{H}_{u\sigma} = B^\top P A W_u = F W_u, \quad \tilde{H}_{u\omega} = B^\top P A W_y = F W_y, \quad \tilde{H}_{uu} = I_m + B^\top P B. \quad (28)$$

The model-free output-feedback Bellman equation is [19, Eq. (2.27)]

$$\tilde{z}_k^\top \tilde{H} \tilde{z}_k = y_k^\top Q_y y_k + u_k^\top u_k + \tilde{z}_{k+1}^\top \tilde{H} \tilde{z}_{k+1}, \quad (29)$$

with  $Q = C^\top Q_y C$ , requiring  $L \geq \tilde{l}(\tilde{l} + 1)/2$  samples. The optimal output-feedback law  $u_k^* = -(\tilde{H}_{uu}^*)^{-1}(\tilde{H}_{u\sigma}^* \sigma_k + \tilde{H}_{u\omega}^* \omega_k)$  is the steady-state equivalent of the absolute regulator  $-K^*x_k$  [19, Thm. 2.5].

### 2.3.2. Output-Feedback Disagreement Reconstruction

The absolute regulator  $-K^*x_i$  learned above drives each agent independently to the origin and does *not* realize the consensus protocol (6), which acts on the inter-agent disagreement  $x_i - x_j$ . The following result shows that this disagreement feedback, together with  $G$ , is recovered directly from the learned blocks (28), yielding a model-free output-feedback realization of (6).

**Proposition 1** (Output-Feedback Disagreement Reconstruction). *Let  $\tilde{H}^*$  be the converged output-feedback  $Q$ -matrix with blocks (28) evaluated at  $P^*$ , and let all agents run identical filters (26). Then the predictor coupling gain is recovered as*

$$G = \tilde{H}_{uu}^* - I_m = B^\top P^* B, \quad (30)$$

and, for any agents  $i, j$  and all  $k \geq n$  under the dead-beat parameterization, the state-disagreement feedback in (6) is reconstructed from the filter states as

$$F(x_i - x_j) = \tilde{H}_{u\sigma}^*(\sigma_i - \sigma_j) + \tilde{H}_{u\omega}^*(\omega_i - \omega_j). \quad (31)$$

*Proof.* The gain (30) is immediate from  $\tilde{H}_{uu}^* = I_m + B^\top P^* B$  in (28). For (31), the dead-beat parameterization (25) applied to agents  $i$  and  $j$  gives, for  $k \geq n$ ,

$$\begin{aligned} x_i &= W_u \sigma_i + W_y \omega_i, \\ x_j &= W_u \sigma_j + W_y \omega_j, \end{aligned}$$

so that, by linearity,

$$x_i - x_j = W_u(\sigma_i - \sigma_j) + W_y(\omega_i - \omega_j).$$

Premultiplying by  $F$  and substituting  $\tilde{H}_{u\sigma}^* = FW_u$  and  $\tilde{H}_{u\omega}^* = FW_y$  from (28),

$$\begin{aligned} F(x_i - x_j) &= FW_u(\sigma_i - \sigma_j) + FW_y(\omega_i - \omega_j) \\ &= \tilde{H}_{u\sigma}^*(\sigma_i - \sigma_j) + \tilde{H}_{u\omega}^*(\omega_i - \omega_j), \end{aligned}$$

which is (31).  $\square$

Consequently, each agent deploys the predictor protocol directly from output data,

$$u_i = -\kappa_i \sum_{j \in \mathcal{N}_i} \left[ \tilde{H}_{u\sigma}^*(\sigma_i - \sigma_j) + \tilde{H}_{u\omega}^*(\omega_i - \omega_j) + G(u_i - u_j) \right], \quad (32)$$

with the implicit coupling resolved by the  $\gamma$ -step iteration (12)–(13) using the reconstructed disagreement in place of  $F(x_i - x_j)$ . In implementation, every agent

applies  $u_i = 0$  during the first  $n$  steps while its filters fill (the dead-beat warm-up), after which the reconstruction (31) is *exact* and the protocol engages. This is the output-feedback counterpart of (6); it differs fundamentally from the absolute regulator  $-\bar{K}^*[\sigma_i^\top \omega_i^\top]^\top = -K^*x_i$  of [19], which does not implement consensus. Fig. 2 demonstrates both halves of Proposition 1 numerically on the systems of Section 3. The left panel shows the reconstruction error  $\|F(x_i - x_j) - [\tilde{H}_{u\sigma}^*(\sigma_i - \sigma_j) + \tilde{H}_{u\omega}^*(\omega_i - \omega_j)]\|$  along an arbitrary two-agent trajectory with learned blocks: it collapses from  $O(10)$  to numerical zero precisely at  $k = n$ , the end of the dead-beat transient. The right panel shows the resulting deployment equivalence: across all three simulation scenarios of Section 3, the output-feedback network trajectories coincide with their state-feedback counterparts to within  $10^{-11}$ , i.e., machine precision for the simulation horizon.

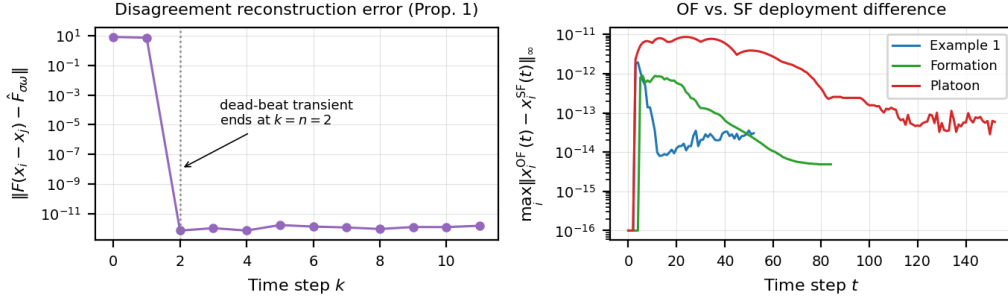


Figure 2: Numerical evidence for Proposition 1. Left: disagreement reconstruction error along a two-agent trajectory (Example 1 system); the error is exactly zero for  $k \geq n = 2$ . Right: maximum deviation between output-feedback and state-feedback network deployments for all three scenarios of Section 3; the deployments coincide to machine precision.

### 2.3.3. Convergence Guarantees

**Theorem 2** (Output Feedback Convergence and Consensus). *Let Assumptions 1–3 hold, with  $(A, \sqrt{Q})$  observable and  $W$  of full row rank. Then: (i)  $\tilde{H}^j \rightarrow \tilde{H}^*$ ; (ii)  $G = \tilde{H}_{uu}^* - I_m = B^\top P^* B$  and the disagreement feedback (31) is recovered; (iii) the protocol (32) achieves (multi-)consensus under  $\kappa_i \bar{r} > 1$ ; (iv) the scheme is immune to exploration noise bias.*

*Proof.* **(i) Convergence.** Substituting the steady-state parameterization  $x_k = W_u \sigma_k + W_y \omega_k$  into the output-feedback Bellman equation (29) and collecting terms with  $W = [W_u \ W_y]$  gives

$$W^\top P^j W = W^\top (Q + (K^j)^\top K^j + (A - BK^j)^\top P^j (A - BK^j)) W.$$

Since  $W$  has full row rank, pre- and post-multiplying by  $(WW^\top)^{-1}W$  and  $W^\top(WW^\top)^{-1}$  cancels  $W$  and recovers the state-feedback Lyapunov equation

$$P^j = Q + (K^j)^\top K^j + (A - BK^j)^\top P^j (A - BK^j),$$

so the iterations coincide with the state-feedback PI/VI recursions and  $P^j \rightarrow P^*$  [19, Thms. 2.6, 2.7]. The reconstructed output-feedback blocks

$$\tilde{H}_{u\sigma}^j = B^\top P^j A W_u, \quad \tilde{H}_{u\omega}^j = B^\top P^j A W_y, \quad \tilde{H}_{uu}^j = I_m + B^\top P^j B,$$

therefore converge to  $\tilde{H}^*$  as  $P^j \rightarrow P^*$ . **(ii) Gain and disagreement recovery.** From the converged blocks,

$$G = \tilde{H}_{uu}^* - I_m = B^\top P^* B, \\ F(x_i - x_j) = \tilde{H}_{u\sigma}^*(\sigma_i - \sigma_j) + \tilde{H}_{u\omega}^*(\omega_i - \omega_j),$$

the second line being the reconstruction of Proposition 1, so the protocol (32) is realized entirely from input-output data. **(iii) Consensus.** With  $\tilde{H}_{u\sigma}^* = B^\top P^* A W_u$  and  $\tilde{H}_{uu}^* = I_m + B^\top P^* B$  for  $P^*$  solving (10), the deployed gains equal (11), so by Remark 1 the closed-loop modes

$$\bar{A}_k = A - B\left(\frac{1}{\bar{\lambda}_k} I_m + G\right)^{-1} F$$

are Schur for every nonzero  $\bar{\lambda}_k$  whenever  $\kappa_i \bar{r} > 1$ , exactly as in Theorem 1(iii). **(iv) Noise immunity.** The extended Bellman equation (29) has the same symmetric structure in  $e_k$  as (22), so the scheme inherits the bias-free property of [19, Thm. 2.8].  $\square$

### 3. Results

We validate the approach on two scenarios adapted from [13]—multi-leader following ( $N=9$ ) and formation control ( $N=6$ ,  $T = 0.1$  s)—together with a robustness study and a vehicle-platoon example. Every scenario is executed under *both* state feedback and output feedback, and validation is performed by direct comparison with the model-based design of [13]: the Q-learned gains are checked for exact numerical agreement with the DARE solution, and the output-feedback deployment is checked for exact agreement with the state-feedback deployment (Proposition 1). All simulations use  $R = I_m$  and VI, and the output-feedback runs employ the  $n$ -step dead-beat filter warm-up of Section 2.3.2.



---

**Algorithm 2** Output Feedback Q-Learning for Consensus Gains
 

---

**Require:** System order  $n$ , input/output dimensions  $m, p$ ; cost  $Q_y \geq 0$ ; tolerance  $\varepsilon$ .

**Ensure:** Consensus gain  $G$  and disagreement-feedback blocks  $\tilde{H}_{u\sigma}^*, \tilde{H}_{u\omega}^*$ .

- 1: Set  $R = I_m, \tilde{l} = mn + pn + m, \tilde{N}_p = \tilde{l}(\tilde{l} + 1)/2$ .
  - 2: Choose companion  $\mathcal{A} \in \mathbb{R}^{n \times n}$  (eigenvalues  $\neq \text{eig}(A)$ ; e.g., all zeros for dead-beat),  $\mathcal{B} = [1 \ 0 \ \dots \ 0]^\top$ .
  - 3: Initialize  $\tilde{H}^0 \in \mathbb{R}^{\tilde{N}_p}, \tilde{K}^0 = 0, j \leftarrow 0; \sigma_0^i = 0 \ (i = 1, \dots, m), \omega_0^i = 0 \ (i = 1, \dots, p)$ .
  - 4: **for**  $k = 0, 1, \dots, L - 1$  **do**
  - 5:   Apply  $u_k = -\tilde{K}^j [\sigma_k^\top \ \omega_k^\top]^\top + e_k$ , measure  $y_k$ ; update filters via (26)
  - 6:   Form  $\tilde{z}_k = [\sigma_k^\top \ \omega_k^\top \ u_k^\top]^\top, \tilde{\tilde{z}}_k \in \mathbb{R}^{\tilde{N}_p}$ , and  $r_k = y_k^\top Q_y y_k + u_k^\top u_k$
  - 7: **end for**
  - 8: **repeat**
  - 9:   — **Least Squares Solve (VI)** —
  - 10:   Build  $\tilde{\Phi} \in \mathbb{R}^{L \times \tilde{N}_p}$ : row  $k = \tilde{\tilde{z}}_k^\top$ ;  $\tilde{\Psi} \in \mathbb{R}^L$ : entry  $k = r_k + (\tilde{H}^j)^\top \tilde{\tilde{z}}_{k+1}$ ; solve  $\tilde{\Phi} \tilde{H}^{j+1} = \tilde{\Psi}$
  - 11:   Reshape  $\tilde{H}^{j+1} \rightarrow \tilde{H}^{j+1}$ ; update  $\tilde{K}^{j+1} = (\tilde{H}_{uu}^{j+1})^{-1} [\tilde{H}_{u\sigma}^{j+1} \ \tilde{H}_{u\omega}^{j+1}]$ ,  $j \leftarrow j + 1$
  - 12: **until**  $\|\tilde{H}^j - \tilde{H}^{j-1}\| < \varepsilon$
  - 13: — **Extract Gains and Deploy** —
  - 14:  $G = \tilde{H}_{uu}^* - I_m$ ; deploy the consensus protocol (32) using the reconstructed disagreement feedback (31) (Proposition 1) after an  $n$ -step filter warm-up.
- 

### 3.1. Example 1: Multi-Leader Following ( $N = 9, \mu = 3$ )

We consider  $N = 9$  agents with sampled double integrator dynamics ( $T = 1$ )

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} T^2/2 \\ T \end{bmatrix} = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad (33)$$

communicating over the digraph in Fig. 3. Nodes 1, 2, and 3 have no in-neighbors and are therefore genuine roots: the graph has  $\mu = 3$  reaches with exclusive parts  $\mathcal{H}_k = \{k\}, k = 1, 2, 3$ , and common part  $C = \{4, \dots, 9\}$ . The coarsest AEP has  $p = 4$  common cells,

$$C_{\mu+1} = \{4, 5, 6\}, \quad C_{\mu+2} = \{7\}, \quad C_{\mu+3} = \{8\}, \quad C_{\mu+4} = \{9\},$$

with mixing vectors  $\gamma^k$  giving asymptotic leader weights  $(\frac{1}{2}, \frac{1}{2}, 0)$  for node 7,  $(0, \frac{1}{2}, \frac{1}{2})$  for node 8,  $(\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$  for node 9, and  $(\frac{1}{8}, \frac{1}{2}, \frac{3}{8})$  for the cell  $\{4, 5, 6\}$ . The

nonzero Laplacian eigenvalues are all equal to 2, so  $\bar{r} = 2$  and the choice  $\kappa_i = 2$  gives  $\kappa_i \bar{r} = 4 > 1$  with strict margin. We set  $Q_y = 1$ , hence  $Q = C^\top Q_y C$ , and  $x_i(0) = i\kappa_2$ . The model-based DARE solution gives

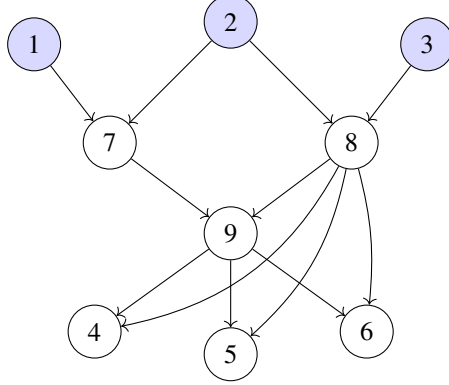


Figure 3: Communication digraph for Example 1 ( $N = 9$ ,  $\mu = 3$ ). Blue nodes are leaders (zero in-degree); each follower cell of the coarsest AEP receives a distinct convex mix of the leader trajectories.

$$P^* = \begin{bmatrix} 2 & 1 \\ 1 & 1.5 \end{bmatrix}, \quad F = \begin{bmatrix} 2 & 4 \end{bmatrix}, \quad G = 3.$$

### 3.1.1. State Feedback Q-Learning Results

Algorithm 1 is applied to a single agent ( $l = 3$ ,  $N_p = 6$ ,  $L = 36$  samples). Fig. 4 shows convergence in 26 iterations to machine precision. Table 3 confirms that the Q-learned gains match the model-based DARE solution to machine precision, verifying exact reproduction of the model-based design. Fig. 5 shows multi-

Table 3: Reproduction verification: Q-learned vs. model-based gains for Example 1.

| Quantity   | Model-based [13] | Q-learned        | Error        |
|------------|------------------|------------------|--------------|
| $P_{11}^*$ | 2.0000           | 2.0000           | $< 10^{-12}$ |
| $P_{12}^*$ | 1.0000           | 1.0000           | $< 10^{-12}$ |
| $P_{22}^*$ | 1.5000           | 1.5000           | $< 10^{-12}$ |
| $F$        | [2.0000, 4.0000] | [2.0000, 4.0000] | $< 10^{-13}$ |
| $G$        | 3.0000           | 3.0000           | $< 10^{-13}$ |

consensus trajectories using centralized ( $\kappa = 2$ ), distributed ( $\kappa = 2$ ,  $\gamma = 2$ ), and standard ( $\kappa = 0.28$  [10]) protocols. The centralized and distributed protocols reach the multi-consensus within  $\sim 8$  steps after the protocol engages, while the standard protocol requires  $\sim 20$  steps with a visibly oscillatory transient. The velocity

State FB Q-Learning Convergence (Example 1)



Figure 4: State feedback Q-learning convergence for Example 1. Left: Bellman residual. Center: H-matrix error. Right: gain error.

components converge exactly to the AEP-predicted levels: 1, 2, 3 for the leaders and 1.5, 2.5, 2.0, 2.25 for the cells  $\{7\}$ ,  $\{8\}$ ,  $\{9\}$ ,  $\{4, 5, 6\}$ , respectively. The Q-learned gains produce trajectories identical to the model-based design, since the learned  $(F, G)$  equal the DARE solution to machine precision.

### 3.1.2. Output Feedback Q-Learning Results

Algorithm 2 uses  $y_k = [1 \ 0]x_k$  with dead-beat parameterization ( $\tilde{l} = 5$ ,  $\tilde{N}_p = 15$ ,  $L = 120$  samples). Fig. 6 shows convergence in 110 iterations, with the learned blocks

$$\tilde{H}_{u\sigma}^* = [5 \ 3], \quad \tilde{H}_{u\omega}^* = [8 \ -6], \quad \tilde{H}_{uu}^* = 4,$$

so that  $G = \tilde{H}_{uu}^* - I_m = 3$  matches the DARE value exactly. Fig. 7 shows the multi-consensus trajectories under the output-feedback deployment (32), with the disagreement reconstructed from the filter states. The trajectories coincide with the state feedback case of Fig. 5 to within  $1.9 \times 10^{-12}$  (Fig. 2, right), which is the quantitative content of Proposition 1: output feedback loses nothing.

### 3.2. Example 2: Formation Control ( $N = 6$ , $T = 0.1$ s)

We consider  $N = 6$  continuous-time double integrator agents over the ring digraph in Fig. 8, which has one reach. Setting  $x_i = \text{col}(\zeta_i - \sigma_i, \dot{\zeta}_i)$ , the discrete-time model with  $T = 0.1$  s has  $n = 4$  states and  $m = 2$  inputs, and position-only measurements  $C = [I_2 \ 0]$  ( $p = 2$ ). We use  $Q_y = 5I_2$ , hence  $Q = C^\top Q_y C$  (Remark 1), for both feedback types so that the state and output feedback designs solve the same LQR problem. The nonzero Laplacian eigenvalues of the directed ring satisfy  $\bar{r} = 1 - \cos(2\pi/6) = 0.5$  *exactly*, so the coupling gain must satisfy  $\kappa > 2$  strictly; we take  $\kappa = 2.5$ , giving  $\kappa\bar{r} = 1.25 > 1$ . The formation is a regular

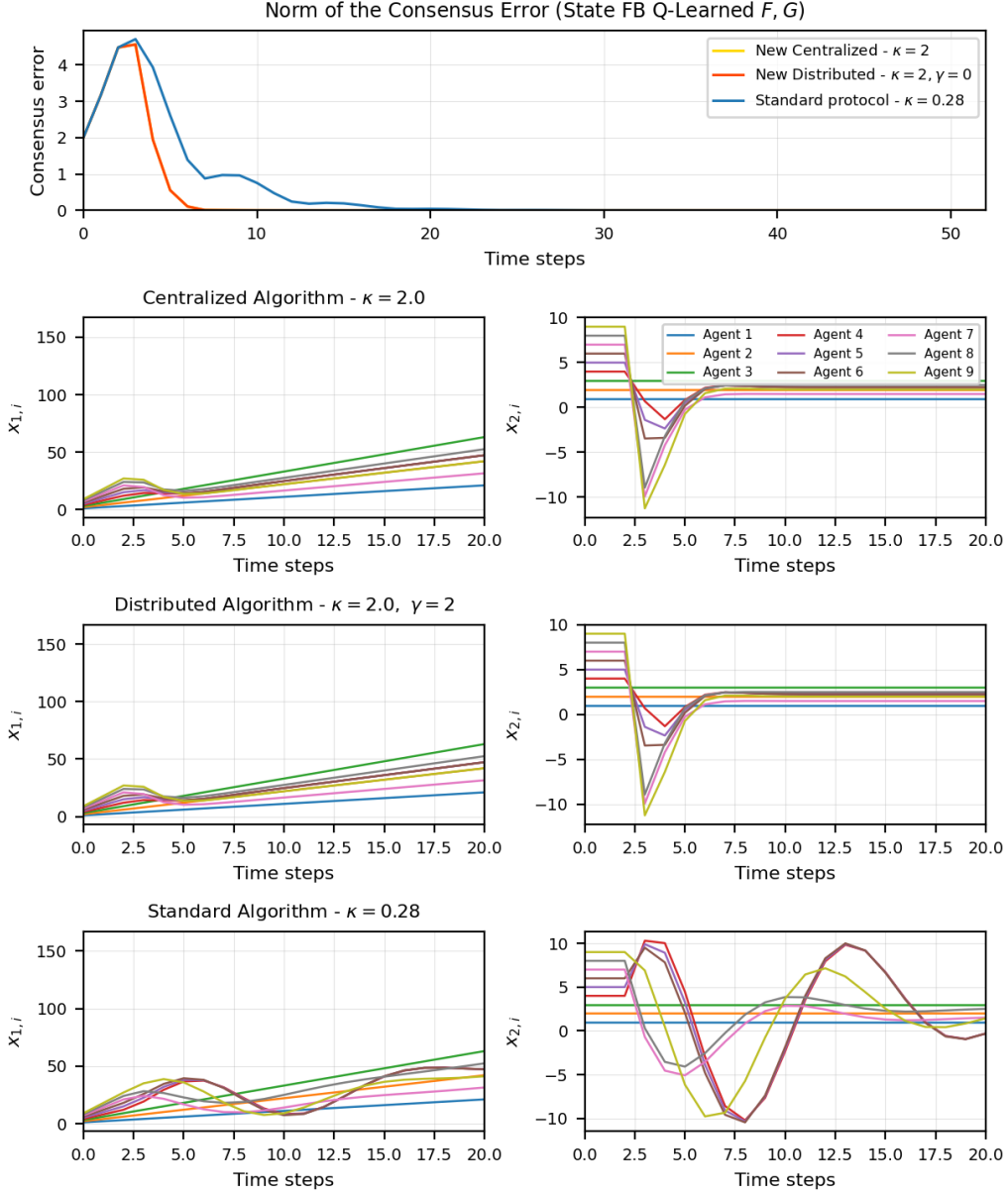


Figure 5: Multi-consensus trajectories with state feedback Q-learned gains for Example 1. Top: consensus error. Rows 2–4: centralized, distributed ( $\gamma = 2$ ), standard protocol. The first  $n = 2$  steps are the zero-input warm-up used for alignment with the output feedback runs.

Output FB Q-Learning Convergence (Example 1)

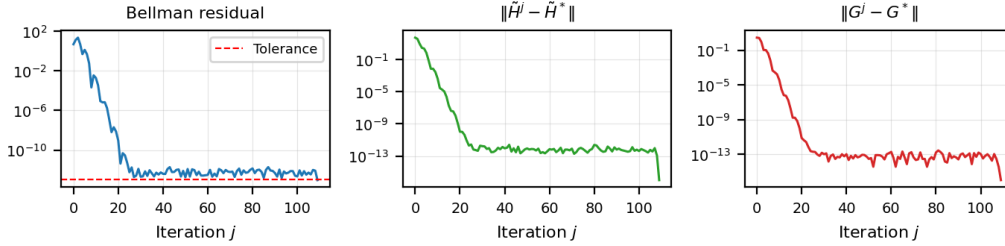


Figure 6: Output feedback Q-learning convergence for Example 1.

hexagon. This scenario represents a practical UAV coordination problem [2, 16]: each agent controls its 2D position and velocity to achieve a desired geometric formation while communicating only with neighbors. The data-driven approach is particularly relevant here because the discrete-time model depends on the sampling period  $T$  through the matrix exponential  $A = \exp(A_c T)$ , and discretization errors are unavoidable in practice. The model-based gains are

$$F = \begin{bmatrix} 2.4853 & 0 & 2.4780 & 0 \\ 0 & 2.4853 & 0 & 2.4780 \end{bmatrix}, \quad G = 0.2354 I_2.$$

### 3.2.1. State Feedback Results

With  $l = 6$ ,  $N_p = 21$ , Fig. 9 shows convergence in 156 iterations with  $\|F - F^*\| < 10^{-12}$ . Fig. 10 shows the agents converging to the hexagonal formation within  $\sim 5$  s under both the centralized protocol and its distributed  $\gamma$ -step approximation with  $\gamma = 35$ , selected empirically so that the distributed input matches the centralized input to below  $10^{-14}$  (Section 4.3).

### 3.2.2. Output Feedback Results

With  $\tilde{l} = mn + pn + m = 18$  and  $\tilde{N}_p = 171$ , output feedback Q-learning converges in 137 iterations with  $\|G - G^*\| < 10^{-12}$  (Fig. 11). Fig. 12 shows the formation trajectories deployed via the reconstructed disagreement (31) using position measurements only: the trajectories coincide with the state feedback case of Fig. 10 to within  $8.8 \times 10^{-13}$  (Fig. 2, right).

### 3.3. Robustness to Parametric Uncertainty

Because the Bellman equations (22) and (29) are built entirely from measured data, the Q-learned gains are matched to the true plant from which the data is

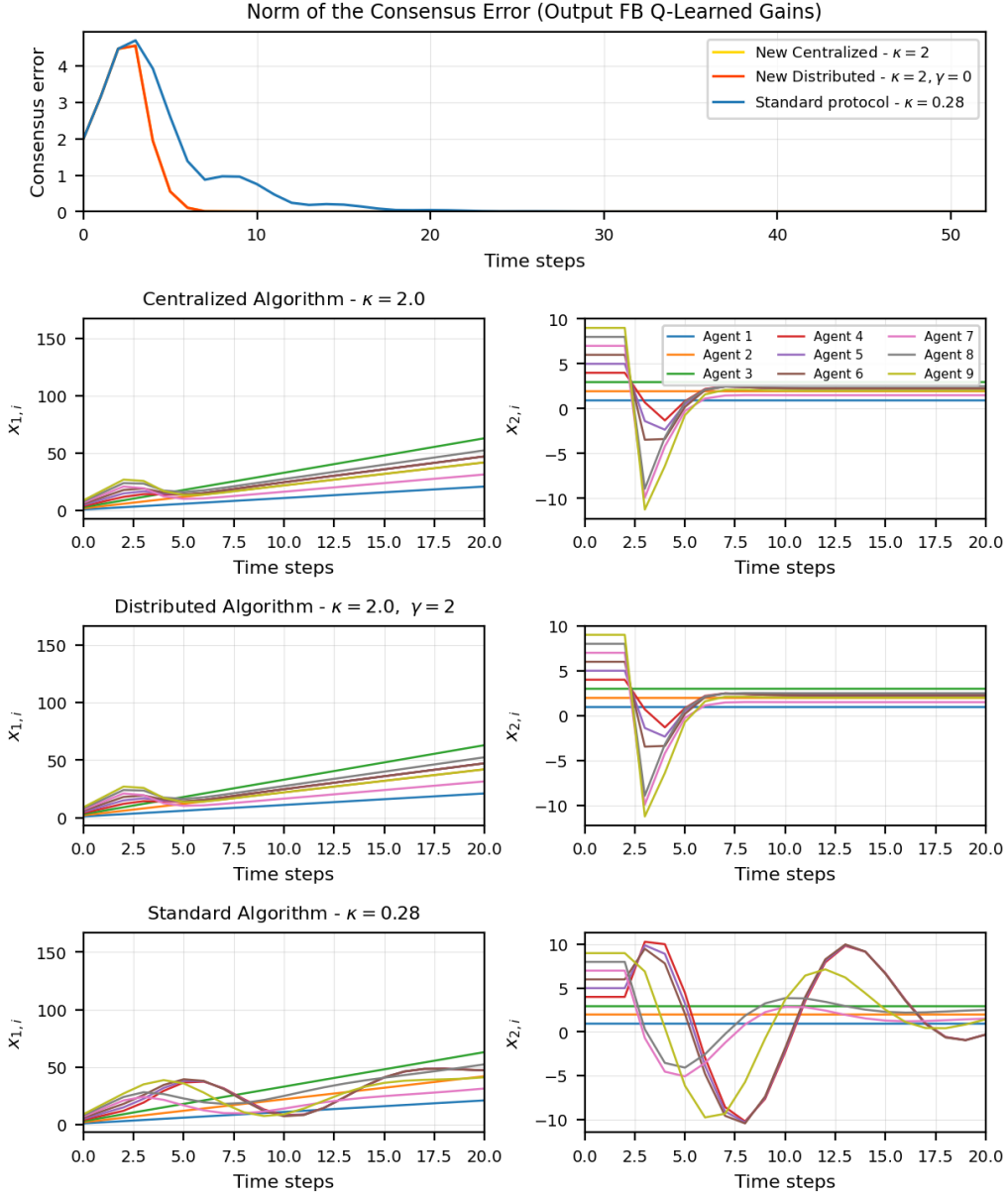


Figure 7: Multi-consensus with output feedback Q-learned gains for Example 1, deployed via the reconstructed disagreement feedback (31). The trajectories coincide with the state feedback case (Fig. 5) to machine precision, validating Proposition 1.

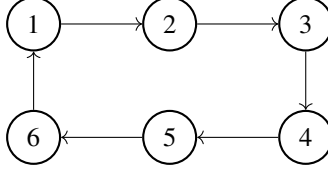


Figure 8: Ring digraph for Example 2 ( $N = 6$ ,  $\mu = 1$ ,  $\bar{r} = 0.5$ ).

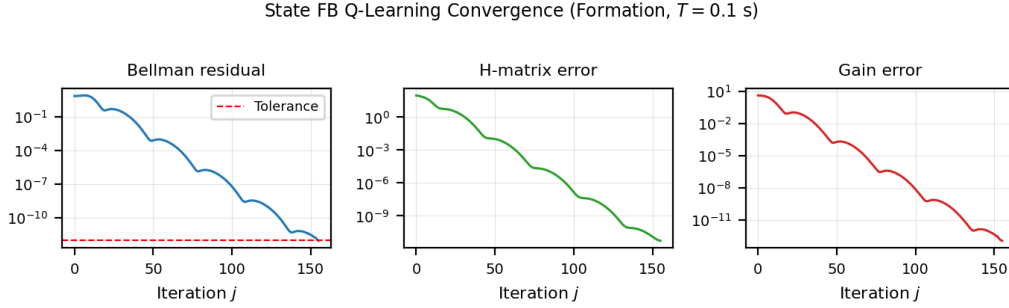


Figure 9: State feedback Q-learning convergence for Example 2 ( $T = 0.1$  s).

collected. To quantify this advantage under model uncertainty, we consider Example 1 with a perturbed system matrix  $A_{\text{true}} = A_{\text{nom}} + \Delta A$ , where  $\Delta A = \delta \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$  represents uncertainty in the (1, 2) entry of the double integrator. The model-based approach computes gains from the nominal  $A_{\text{nom}}$  via the DARE, while Q-learning extracts gains from data collected on the true (perturbed) system. Since the state and output feedback algorithms recover the identical  $(F, G)$  (Theorems 1 and 2), a single Q-learned series represents both. Fig. 13 presents four analyses. Panel (a) shows that, in this example, the maximum closed-loop spectral radius  $\max_k \rho(\bar{A}_k)$  for the model-based gains increases monotonically with the perturbation magnitude, approaching the stability boundary  $\rho = 1$ , while the Q-learned spectral radius *decreases* over the tested range. At  $\delta = 1.0$ , the model-based margin is  $1 - 0.832 = 0.168$  versus  $1 - 0.196 = 0.804$  for Q-learning—a  $4.8\times$  improvement. Panel (b) shows that the relative LQR cost penalty of model-based gains grows to 114% at  $\delta = 1.0$  ( $J_{\text{nom}}/J_{\text{QL}} = 2.14$ ). Panel (c) shows the velocity trajectories at  $\delta = 0.3$ : the Q-learned gains (dashed) achieve tighter consensus than the model-based gains (solid), consistent with the spectral radius difference. Panel (d) compares the gain values at  $\delta = 0.5$ : the Q-learned gains (2.338, 6.803, 4.465) match the true DARE solution to machine precision, while the nominal model gains (2, 4, 3) deviate significantly.

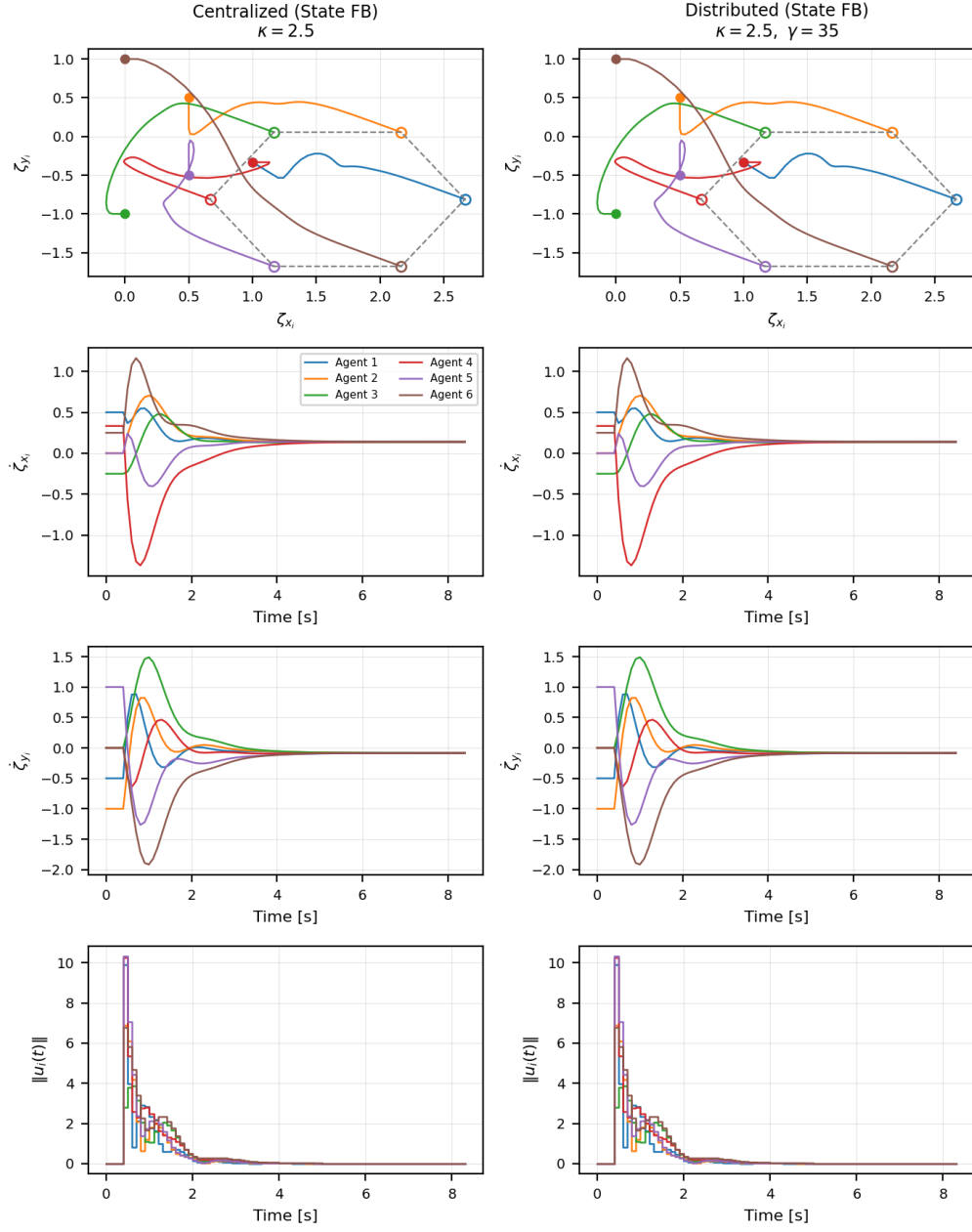


Figure 10: Formation control with state feedback Q-learned gains for Example 2 ( $T = 0.1$  s,  $\kappa = 2.5$ ). Left column: centralized; right column: distributed ( $\gamma = 35$ ). Agents converge to the hexagonal formation within  $\sim 5$  s.



Output FB Q-Learning Convergence (Formation,  $T = 0.1$  s)

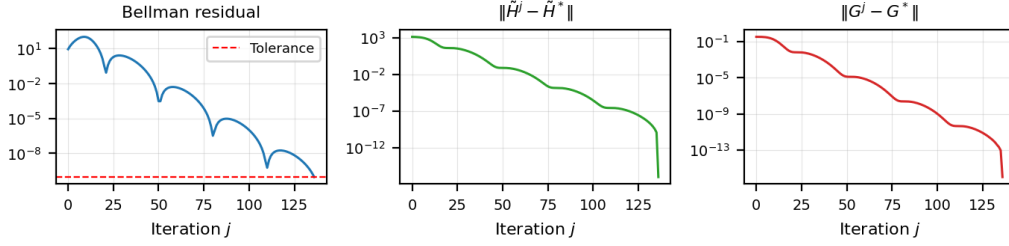


Figure 11: Output feedback Q-learning convergence for Example 2 ( $T = 0.1$  s).

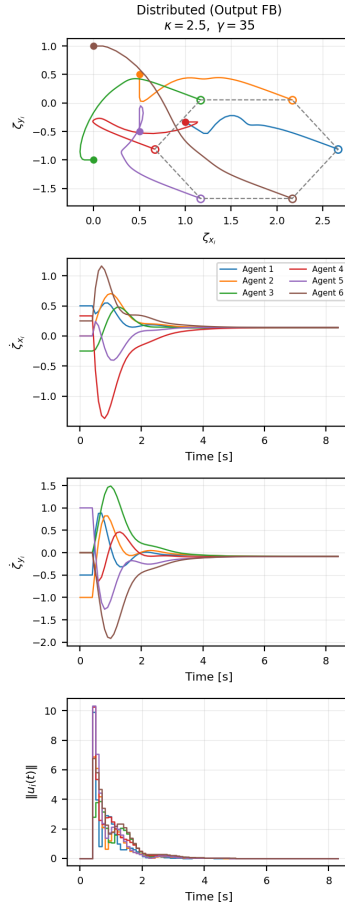


Figure 12: Formation control with output feedback Q-learned gains for Example 2, using position measurements only ( $C = [I_2 \ 0]$ ) and the reconstructed disagreement feedback (31). The trajectories coincide with Fig. 10 to machine precision.

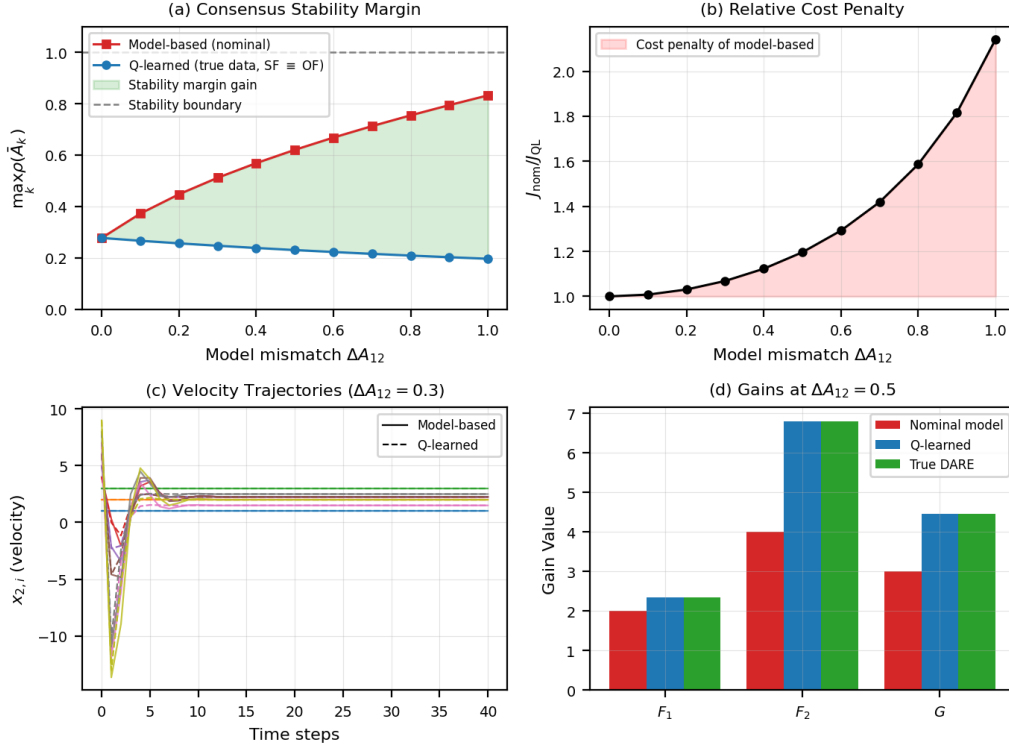


Figure 13: Robustness analysis under parametric uncertainty  $\Delta A_{12}$ . Top-left: stability margin (the Q-learned series represents both state and output feedback, which learn identical gains). Top-right: cost penalty. Bottom-left: velocity trajectories at  $\delta = 0.3$  (solid = model-based, dashed = Q-learned). Bottom-right: gain comparison at  $\delta = 0.5$ .

### 3.4. Example 3: Vehicle Platoon with Aerodynamic Drag

To demonstrate the practicality of the proposed framework, we consider a vehicle platoon control problem—a representative application in intelligent transportation systems. A string of  $N = 5$  vehicles with aerodynamic drag must achieve velocity consensus while maintaining a desired inter-vehicle spacing of  $d_{\text{ref}} = 10$  m. Each vehicle has the discrete-time dynamics

$$A = \begin{bmatrix} 1 & T \\ 0 & 1 - bT \end{bmatrix}, \quad B = \begin{bmatrix} T^2/2 \\ T \end{bmatrix}, \quad (34)$$

where  $T = 0.1$  s is the sampling period and  $b = 0.1 \text{ s}^{-1}$  is the drag coefficient. Unlike the double integrator, this system includes velocity-dependent dissipation ( $bT = 0.01$ ), making it a more realistic model of longitudinal vehicle dynamics. The state is  $x_i = [\zeta_i + d_{\text{ref}}, \dot{\zeta}_i]^T$  (spacing coordinate and velocity), the measurement is position only,  $C = [1 \ 0]$  with  $Q_y = 1$ , and the communication graph is

a directed string  $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$  with  $\bar{r} = 1$  and  $\kappa = 2$ . The model-based gains are  $F = [1.0684 \ 1.4555]$ ,  $G = 0.1416$ . Under state feedback, Algorithm 1 converges in 234 iterations with  $\|F - F^*\| < 10^{-13}$  (Fig. 14, top row); under output feedback, Algorithm 2 converges in 208 iterations with  $\|G - G^*\| < 10^{-12}$  (Fig. 14, bottom row). Fig. 15 shows the platoon behavior under the state feedback deployment ( $\gamma = 6$ ): vehicles achieve the desired 10 m spacing with velocity consensus, and control inputs remain smooth and bounded. Fig. 16 shows the same scenario under the output feedback deployment using position sensors only: the trajectories coincide with the state feedback case to within  $8.6 \times 10^{-12}$  (Fig. 2, right), confirming that the platoon can be controlled without velocity measurement. The data-driven approach is particularly advantageous here because the drag coefficient  $b$  may vary with speed and environmental conditions, making the exact discrete-time model uncertain.

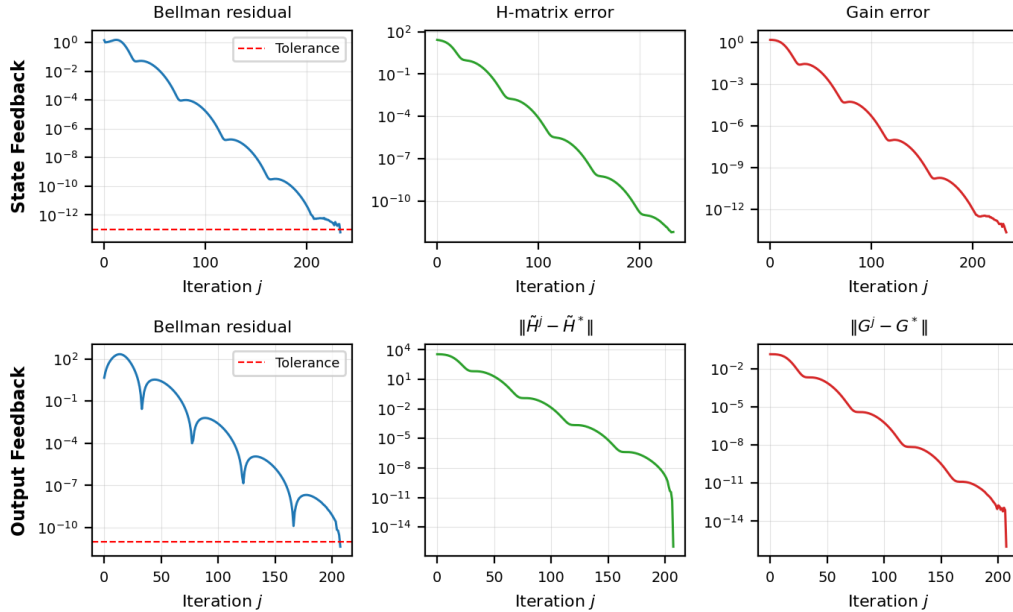


Figure 14: Q-learning convergence for the platoon example. Top row: state feedback. Bottom row: output feedback.

### 3.5. Summary of Results

Table 4 summarizes convergence across all examples. Every example is validated under both feedback types, and in every case the output feedback deployment reproduces the state feedback deployment to machine precision.

**Vehicle Platoon with Aerodynamic Drag ( $b = 0.1 \text{ 1/s}$ ,  $T = 0.1 \text{ s}$ )**  
**State FB Q-Learned Predictor-Based Consensus Protocol**

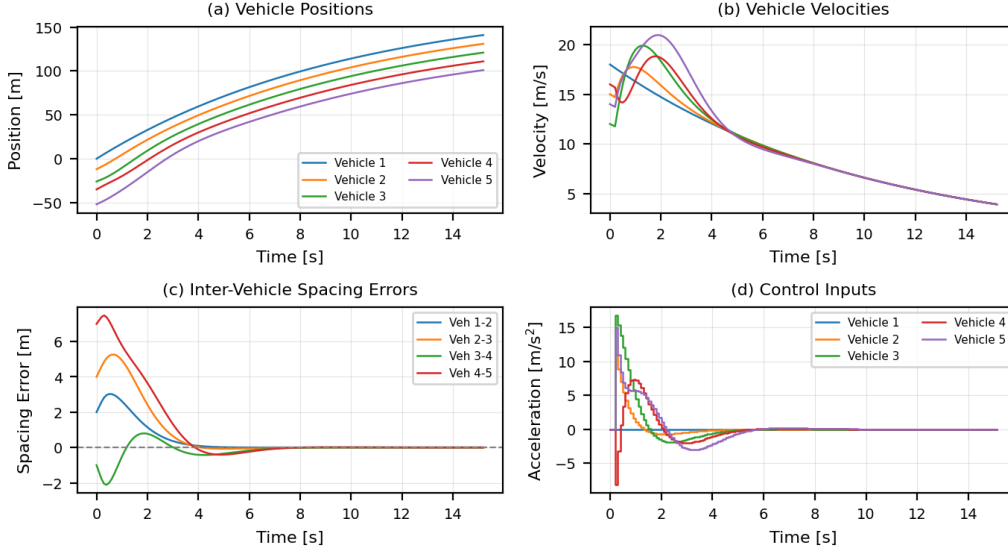


Figure 15: Vehicle platoon control with state feedback Q-learned gains ( $N = 5$ ,  $b = 0.1 \text{ s}^{-1}$ ,  $T = 0.1 \text{ s}$ ). (a) positions, (b) velocities, (c) spacing errors, (d) control inputs.

Table 4: Summary of Q-learning convergence across examples.

| Example                           | $n$ | $m$ | Method               | Iterations | Final accuracy           |
|-----------------------------------|-----|-----|----------------------|------------|--------------------------|
| 1 ( $N=9$ , $\mu=3$ )             | 2   | 1   | State FB             | 26         | $\ F - F^*\  < 10^{-13}$ |
|                                   |     |     | Output FB            | 110        | $\ G - G^*\  < 10^{-12}$ |
|                                   |     |     | OF vs. SF deployment |            | $< 1.9 \times 10^{-12}$  |
| 2 (Formation, $T=0.1 \text{ s}$ ) | 4   | 2   | State FB             | 156        | $\ F - F^*\  < 10^{-12}$ |
|                                   |     |     | Output FB            | 137        | $\ G - G^*\  < 10^{-12}$ |
|                                   |     |     | OF vs. SF deployment |            | $< 8.8 \times 10^{-13}$  |
| 3 (Platoon)                       | 2   | 1   | State FB             | 234        | $\ F - F^*\  < 10^{-13}$ |
|                                   |     |     | Output FB            | 208        | $\ G - G^*\  < 10^{-12}$ |
|                                   |     |     | OF vs. SF deployment |            | $< 8.6 \times 10^{-12}$  |

**Vehicle Platoon with Aerodynamic Drag ( $b = 0.1 \text{ 1/s}$ ,  $T = 0.1 \text{ s}$ )**  
**Output FB Q-Learned Predictor-Based Consensus Protocol**

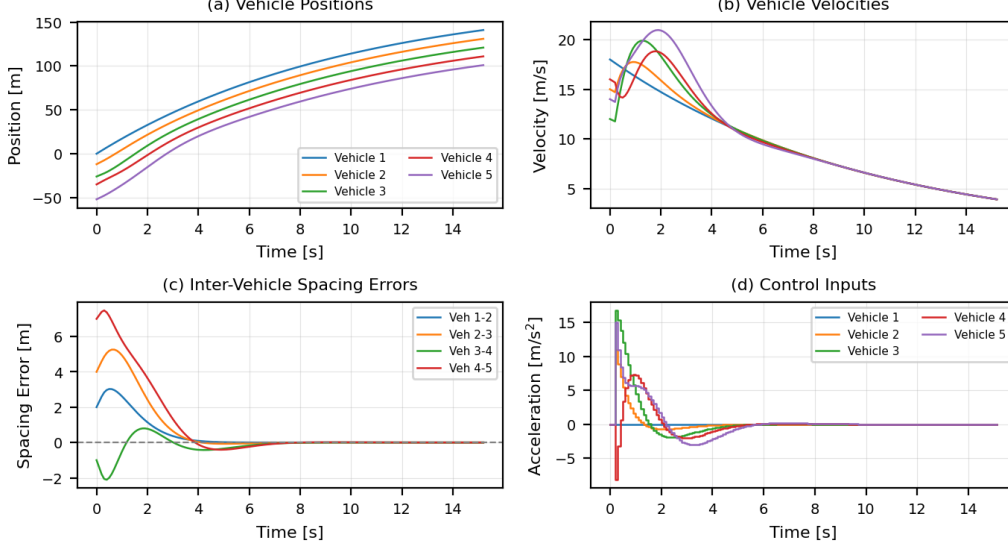


Figure 16: Vehicle platoon control with output feedback Q-learned gains, using position sensors only. The trajectories coincide with Fig. 15 to machine precision.

## 4. Discussion

This section interprets the results and analyzes the practical properties of the proposed framework: computational cost, scalability with network size, selection of the distributed iteration depth, communication requirements, the nature of the observed robustness advantage, and limitations.

### 4.1. Computational Complexity

Each iteration of Algorithm 1 requires solving an  $l(l+1)/2$ -dimensional least-squares problem, with computational cost  $O(L \cdot l^2 + l^6)$  per iteration and total cost  $O(J \cdot l^6)$ , where  $J$  is the number of iterations to convergence and  $l = n + m$ . For comparison, the model-based DARE is solved via the Schur decomposition in  $O(n^3)$ . While Q-learning is more expensive per solve, it avoids the cost and error of system identification, which can be substantial for high-dimensional or poorly conditioned systems. The output-feedback algorithm has the same structure with  $\tilde{l} = mn + pn + m$  in place of  $l$ ; the larger regression dimension ( $\tilde{N}_p = 171$  vs.  $N_p = 21$  in Example 2) increases the per-iteration cost and the required data

length, though the iteration counts are comparable (137 vs. 156 in Example 2, Table 4).

#### 4.2. Scalability with Network Size

A key practical property, confirmed across all examples, is that the learning cost is *independent* of the network size  $N$ . Since the DARE depends only on  $(A, B, Q)$  and not on the communication graph, Algorithms 1–2 are executed once on a single agent, and the learned gains are deployed across all  $N$  agents with their respective coupling gains  $\kappa_i$ . Whether the network contains 5 or 500 agents, the learning phase is identical. This contrasts with distributed RL methods such as [20], where each agent must learn concurrently and the learning cost scales with  $N$ .

#### 4.3. Model-Free Selection of the Distributed Iteration Depth $\gamma$

The minimum number of distributed consensus steps  $\gamma_{\min}$  from [13, Thm. 3.4] depends on  $\rho(G)$  and  $\rho(A - BK^*)$ . The spectral radius  $\rho(G) = \rho(H_{uu}^* - I_m)$  is directly available from the converged H-matrix without any model knowledge. Quantities involving the closed-loop matrix  $A - BK^*$ , however, are not directly available from  $H^*$  alone. In a fully data-driven implementation, one may collect closed-loop data under the converged policy  $K^*$  and estimate

$$\hat{A}_{\text{cl}} = X^+ X^\dagger, \quad (35)$$

where  $X = [x_0, \dots, x_{T-1}]$ ,  $X^+ = [x_1, \dots, x_T]$ , and  $X^\dagger$  denotes the pseudoinverse; then  $\rho(A - BK^*)$  is approximated by  $\rho(\hat{A}_{\text{cl}})$ . Alternatively,  $\gamma$  can be selected empirically by increasing the number of inner consensus iterations until the distributed input  $v_i(t, \gamma)$  is sufficiently close to the centralized input (8), which is the procedure used in Example 2:  $\gamma = 20$  already gives a state deviation below  $1.2 \times 10^{-9}$ , and  $\gamma = 35$  reaches numerical precision ( $7 \times 10^{-15}$ ).

#### 4.4. Communication Requirements

The distributed  $\gamma$ -step algorithm (12)–(13) requires  $\gamma$  communication rounds per sampling period  $T$ , introducing a trade-off: larger  $\gamma$  improves the approximation of the centralized solution (8) but increases the communication burden. In practice, communication latency constrains  $\gamma \leq T/\tau_{\text{comm}}$ , where  $\tau_{\text{comm}}$  is the per-round communication delay. Combining the predictor-based protocol with event-triggered communication strategies is a promising direction to reduce this overhead while preserving consensus guarantees. Note that this requirement is inherited from the protocol of [13] and is not introduced by the learning layer: once learned, the gains impose no additional communication.

#### 4.5. Interpretation of the Robustness Results

The robustness study of Section 3.3 demonstrates *exactness on the true plant*: when data is drawn from the true (perturbed) system, Q-learning recovers the DARE solution of that system to machine precision, whereas the model-based design remains anchored to the nominal model and its stability margin erodes as the perturbation grows. We emphasize that this is a property of data-driven exactness rather than a formal robustness margin; the observed  $4.8\times$  margin improvement at  $\delta = 1.0$  and the 114% cost penalty of the nominal design quantify the practical consequence of this distinction in the example considered. The insight transfers directly to applications such as the vehicle platoon of Section 3.4, where physical parameters (e.g., the drag coefficient) vary with operating conditions and an exact nominal model is unavailable.

#### 4.6. Limitations

Three limitations delineate the scope of the present results. First, the framework assumes identical linear agents; heterogeneous or nonlinear dynamics would break the single-agent learning architecture and the quadratic Q-function structure, respectively. Second, the convergence guarantees assume noise-free state or output *measurements*; exploration noise in the input is provably bias-free (Theorems 1 and 2), but measurement noise would require an instrumental-variable or total-least-squares treatment. Third, the excitation (rank) condition on the regression matrix must be verified in practice, which constrains the choice of probing signals during the learning phase.

### 5. Conclusions

This paper presented data-driven Q-learning algorithms for learning the gains of a predictor-based consensus protocol for discrete-time linear MAS. By exploiting the DARE–LQR structural equivalence, the state feedback algorithm extracts both consensus gains from single-agent data, and the output feedback extension—building on the input-output Q-learning of [19]—reconstructs the inter-agent disagreement feedback required by the protocol rather than the absolute regulator. The learned gains exactly satisfy the DARE, so all consensus and multi-consensus guarantees of [13], including arbitrary convergence rate, are inherited; the stability condition was additionally extended to the positive semi-definite weights that output feedback necessitates. This exactness was validated numerically to machine precision against the model-based benchmark, with every scenario executed under both feedback types and the two deployments shown to coincide to

machine precision; the robustness study showed superior stability margins under parametric uncertainty, and a vehicle platoon example confirmed practical applicability with position-only sensing. The novelty of the work lies in the linkage between data-driven LQR reinforcement learning and predictor-based discrete-time consensus—two established areas not previously connected—and its impact is to extend a consensus design with arbitrary convergence rate to settings where the agent model is unknown. Future directions include heterogeneous MAS (requiring distributed Q-learning), time-varying topologies with event-triggered communication, nonlinear extensions via Koopman lifting [23], and safety enforcement through control barrier functions [21].

### **Declaration of Competing Interest**

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### **Funding**

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

### **Data Availability**

The simulation scripts that support the findings of this study are available from the corresponding author upon reasonable request.

### **CRedit Authorship Contribution Statement**

**Md Nur-A-Adam Dony:** Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Writing – original draft, Visualization. **Syed Ali Asad Rizvi:** Conceptualization, Methodology, Supervision, Writing – review & editing.

### **References**

- [1] B.A. Francis, M. Maggiore, Flocking and Rendezvous in Distributed Robotics, Springer, 2016.



- [2] M.N.A. Dony, W. Dong, Distributed robust formation flying and attitude synchronization of spacecraft, *J. Aerosp. Eng.* 34 (2021) 04021002.
- [3] W. Dong, M.N.A. Dony, M. Rafat, Tracking control of vertical taking-off and landing vehicles with parametric uncertainty, *Int. J. Adapt. Control Signal Process.* 34 (2020) 1600–1612.
- [4] F. Dörfler, F. Bullo, Synchronization in complex networks of phase oscillators: A survey, *Automatica* 50 (6) (2014) 1539–1564.
- [5] Z. Li, Z. Duan, G. Chen, L. Huang, Consensus of multiagent systems and synchronization of complex networks: A unified viewpoint, *IEEE Trans. Circuits Syst. I Reg. Papers* 57 (1) (2010) 213–224.
- [6] G. Battistelli, L. Chisci, G. Mugnai, A. Farina, A. Graziano, Consensus-based linear and nonlinear filtering, *IEEE Trans. Autom. Control* 60 (5) (2015) 1410–1415.
- [7] A.V. Proskurnikov, R. Tempo, A tutorial on modeling and analysis of dynamic social networks. Part I, *Annu. Rev. Control* 43 (2017) 65–79.
- [8] Z. Li, G. Wen, Z. Duan, W. Ren, Designing fully distributed consensus protocols for linear multi-agent systems with directed graphs, *IEEE Trans. Autom. Control* 60 (4) (2015) 1152–1157.
- [9] Z. Liu, A. Saberi, A.A. Stoorvogel, D. Nojavanzadeh, Scale-free collaborative protocol design for state and regulated state synchronization of multi-agent systems with arbitrary fast convergence, *J. Franklin Inst.* 358 (9) (2021) 4864–4882.
- [10] Q. Ma, F.L. Lewis, S. Xu, Cooperative containment of discrete-time linear multi-agent systems, *Int. J. Robust Nonlinear Control* 25 (7) (2015) 1007–1018.
- [11] W. Ren, Y. Cao, *Distributed Coordination of Multi-Agent Networks*, Springer, 2010.
- [12] L. Xiao, S. Boyd, Fast linear iterations for distributed averaging, *Syst. Control Lett.* 53 (1) (2004) 65–78.

- [13] F. Cacace, M. Mattioni, S. Monaco, D. Normand-Cyrot, Consensus and multi-consensus for discrete-time LTI systems, *Automatica* 166 (2024) 111718.
- [14] S. Monaco, L. Ricciardi Celsi, On multi-consensus and almost equitable graph partitions, *Automatica* 103 (2019) 53–61.
- [15] M.N.A. Dony, M. Rafat, W. Dong, Distributed command filtered robust tracking control of wave-adaptive modular vessel with uncertainty, in: *Proc. 2020 American Control Conference (ACC)*, Denver, CO, 2020, pp. 2118–2123.
- [16] M.N.A. Dony, Robust Distributed Formation Control of UAVs with Higher-Order Dynamics, Master’s thesis, The University of Texas Rio Grande Valley, 2021.
- [17] F.L. Lewis, D. Vrabie, Reinforcement learning and adaptive dynamic programming for feedback control, *IEEE Circuits Syst. Mag.* 9 (3) (2009) 32–50.
- [18] S.J. Bradtke, B.E. Ydstie, A.G. Barto, Adaptive linear quadratic control using policy iteration, in: *Proc. American Control Conference*, 1994, pp. 3475–3479.
- [19] S.A.A. Rizvi, Z. Lin, Output Feedback Reinforcement Learning Control for Linear Systems, *Control Engineering Series*, Springer Nature, 2022.
- [20] H. Zhang, F.L. Lewis, A. Das, Optimal design for synchronization of cooperative systems: State feedback, observer and output feedback, *IEEE Trans. Autom. Control* 56 (8) (2011) 1948–1952.
- [21] M.N.A. Dony, M.E. Khallil, Adaptive reinforcement learning with control barrier functions for safe state avoidance in discrete event systems, in: *Proc. 8th IEOM Bangladesh Int. Conf.*, Dhaka, Bangladesh, 2025.
- [22] M.N.A. Dony, B. Arhin, Data-driven distributed battery state-of-charge balancing using neural network-based open circuit voltage learning, *IFAC J. Syst. Control* 36 (2026) 100415.
- [23] M.N.A. Dony, Integrating Koopman theory and Lyapunov stability for enhanced model predictive control in nonlinear systems, *ResearchGate preprint*, 2024. <https://doi.org/10.13140/RG.2.2.17950.31046>.

- [24] J.S. Caughman, J.J.P. Veerman, Kernels of directed graph Laplacians, *Electron. J. Combin.* 13 (1) (2006) 39.
- [25] G.A. Hwer, An iterative technique for the computation of the steady state gains for the discrete optimal regulator, *IEEE Trans. Autom. Control* 16 (4) (1971) 382–384.